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Automated CO₂ Risk Detection and Mitigation through Real-Time Data Integration

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Abstract

The effective monitoring of CO₂ emissions and the implementation of robust risk mitigation strategies are pivotal in the context of carbon capture, utilization, and storage (CCUS) systems. This paper introduces an advanced, integrated framework for the real-time detection, assessment, and mitigation of CO₂-related risks, leveraging a comprehensive risk management dataset. This dataset includes detailed attributes such as hazard types, mitigation protocols, remediation procedures, impact assessments, geospatial data, control measures, and the specific rules that trigger risk events. By systematically ingesting this data into a centralized datastore, we enable a dynamic, real-time risk management interface.

The proposed system incorporates real-time field sensor data, continuously monitoring parameters such as CO₂ concentration, pressure, and other relevant environmental variables. Through seamless integration, the system autonomously identifies risk events by detecting deviations from predefined safety thresholds, such as CO₂ leaks or equipment malfunctions. Once a risk is detected, the system generates automatic alerts and initiates a cascade of predefined actions, which include immediate risk identification, severity evaluation, and automatic execution of remediation plans.

This integrated approach enhances operational resilience by providing continuous situational awareness, enabling a proactive risk response without the need for manual intervention. Upon risk detection, the system autonomously triggers the corresponding mitigation strategy, ensuring that the appropriate remedial actions are promptly undertaken. The closed-loop design of the system allows for ongoing risk assessment, real-time updates, and agile responses, all of which contribute to minimizing the operational and environmental impacts associated with CO₂ emissions.

The proposed framework exemplifies a transformative solution for CO₂ monitoring and risk mitigation, facilitating safer and more efficient CCUS operations. By harnessing the power of real-time sensor data, advanced data analytics, and automated decision-making, the system ensures swift risk identification, assessment, and mitigation. Ultimately, this research underscores the potential of an entirely automated risk management ecosystem that integrates historical risk intelligence with real-time operational data, thereby safeguarding CCUS infrastructure and ensuring regulatory compliance.

Introduction

The global imperative to mitigate climate change has intensified the focus on technologies that can effectively manage greenhouse gas emissions, particularly carbon dioxide (CO₂). Among these, Carbon Capture, Utilization, and Storage (CCUS) systems have emerged as a cornerstone of sustainable industrial practices, offering a pathway to significantly reduce atmospheric CO₂ concentrations. However, the deployment and operation of CCUS infrastructure present complex challenges, especially in terms of real-time monitoring, risk detection, and mitigation. The potential for CO₂ leaks, equipment failures, and environmental hazards necessitates a robust, intelligent framework that can ensure operational safety, regulatory compliance, and environmental stewardship.

Traditional risk management approaches in CCUS systems often rely on periodic assessments, manual interventions, and siloed data sources, which can delay response times and increase the likelihood of adverse events. These limitations underscore the need for a transformative shift toward integrated, automated, and data-driven risk management ecosystems. In this context, the convergence of real-time sensor technologies, advanced data analytics, and intelligent automation offers a promising avenue to enhance the resilience and responsiveness of CCUS operations.

This paper introduces a novel, integrated framework designed to address these challenges by enabling real-time detection, assessment, and mitigation of CO₂-related risks. At the heart of this framework lies a comprehensive risk management dataset, encompassing a wide array of attributes such as hazard types, mitigation protocols, remediation procedures, impact assessments, geospatial data, control measures, and rule-based triggers for risk events. By systematically ingesting this data into a centralized datastore, the system facilitates a dynamic and continuously updated risk management interface that supports proactive decision-making.

The proposed system leverages real-time field sensor data, continuously monitoring critical environmental parameters including CO₂ concentration, pressure, and temperature. These sensors serve as the eyes and ears of the system, providing granular visibility into operational conditions and enabling the early identification of anomalies. Through seamless integration with the risk management datastore, the system autonomously detects deviations from predefined safety thresholds such as CO₂ leaks, pressure surges, or equipment malfunctions and initiates a cascade of automated responses.

Upon detection of a risk event, the system generates instantaneous alerts and executes a series of predefined actions, including risk identification, severity evaluation, and the automatic deployment of remediation plans. This closed-loop design ensures that mitigation strategies are not only identified but also executed without delay, minimizing the potential for human error and enhancing the overall agility of the response mechanism. The system's ability to continuously reassess risks and update its response protocols in real time further strengthens its capacity to adapt to evolving operational conditions.

One of the key innovations of this framework is its emphasis on autonomous decision-making, which eliminates the need for manual intervention in critical scenarios. By integrating historical risk intelligence with live operational data, the system creates a feedback loop that supports continuous learning and optimization. This capability is particularly valuable in complex CCUS environments, where the interplay of geological, mechanical, and environmental factors can produce unpredictable risk profiles.

Moreover, the framework's modular architecture allows for scalability and customization, making it suitable for a wide range of CCUS applications from large-scale industrial facilities to localized pilot projects. Its compatibility with existing sensor networks and data platforms ensures ease of integration, while its extensible rule engine enables the incorporation of domain-specific safety protocols and regulatory requirements.

In addition to enhancing operational safety, the proposed system contributes to regulatory compliance and stakeholder transparency. By maintaining a detailed log of risk events, mitigation actions, and system responses, it provides a verifiable audit trail that can support environmental reporting, compliance

verification, and public communication. This feature is particularly important in jurisdictions with stringent environmental regulations and high public scrutiny of carbon management practices.

The significance of this research lies not only in its technical contributions but also in its potential to reshape the paradigm of risk management in CCUS systems. By demonstrating the feasibility and effectiveness of a fully automated, data-driven risk mitigation framework, this work paves the way for safer, more efficient, and more sustainable carbon management operations. It highlights the critical role of real-time data integration, intelligent automation, and adaptive response mechanisms in addressing the multifaceted challenges of CO₂ emissions monitoring and control.

In the sections that follow, we delve into the architectural design of the system, the structure and semantics of the risk management dataset, the integration of sensor data streams, and the logic underpinning the automated response engine. We also present case studies and performance evaluations that illustrate the system's capabilities in real-world scenarios. Through this comprehensive exploration, we aim to provide a blueprint for the next generation of CCUS risk management systems one that is intelligent, responsive, and resilient by design.

Literature Survey

The increasing urgency to mitigate climate change has positioned Carbon Capture, Utilization, and Storage (CCUS) as a critical technology in the global decarbonization strategy. While the core components of CCUS have been extensively studied, the real-time monitoring and risk mitigation aspects remain areas of active research and development.

Foundational Work in CCUS Risk Management

Early literature on CCUS risk management primarily focused on geological storage risks, such as caprock integrity, induced seismicity, and long-term CO₂ leakage. These studies emphasized the importance of site characterization, modelling, and periodic monitoring to ensure containment. However, these approaches often lacked the temporal resolution and automation needed for real-time operational decision-making.

Sensor-Based Monitoring and MMV Protocols

The Measurement, Monitoring, and Verification (MMV) frameworks developed by regulatory bodies and research institutions have laid the groundwork for sensor-based monitoring in CCUS. These frameworks typically involve a combination of in-situ sensors, satellite imaging, and geophysical surveys to track CO₂ plume migration and detect anomalies. While effective for long-term assurance, these systems are often retrospective and manually intensive, limiting their utility in dynamic operational environments.

Integration of Real-Time Data Streams

Recent advancements have focused on integrating real-time sensor data into CCUS operations. These include continuous monitoring of CO₂ concentration, pressure, temperature, and flow rates using fiber-optic sensors, gas analysers, and distributed acoustic sensing (DAS) systems. However, the literature reveals a gap in systematic frameworks that can not only collect but also interpret and act on this data autonomously.

Machine Learning and Predictive Analytics in CCUS

The application of machine learning (ML) in CCUS has gained traction to enhance predictive capabilities and automate anomaly detection. Du et al. (2025) provide a comprehensive review of ML applications across the CCUS value chain, highlighting use cases such as leak detection, reservoir behaviour prediction, and process optimization. Despite these advances, most implementations remain experimental or domain-specific, lacking generalizable architectures for real-time risk mitigation.

Automated Risk Mitigation Systems

There is a growing interest in closed-loop systems that combine real-time monitoring with automated response mechanisms. These systems aim to reduce human intervention by using predefined rules, severity thresholds, and remediation protocols to trigger alerts and corrective actions. However, the literature indicates that such systems are still in their infancy, with limited examples of fully integrated, end-to-end risk management platforms that can operate autonomously in live CCUS environments.

Data-Centric Risk Intelligence

Another emerging trend is the use of comprehensive risk management datasets that include historical incident data, mitigation strategies, and geospatial attributes. These datasets enable context-aware decision-making and support the development of adaptive risk models. Yet, the integration of such datasets with real-time monitoring systems remains a technical and organizational challenge, often hindered by data silos and interoperability issues.

Regulatory and Operational Implications

From a regulatory standpoint, real-time risk mitigation systems offer the potential to streamline compliance reporting and enhance transparency. They can provide auditable logs of risk events and responses, which are valuable for both internal governance and external oversight. However, the literature also cautions against over-reliance on automation without robust fail-safe mechanisms and human-in-the-loop controls.

This survey highlights the need for a holistic, integrated framework that bridges the gap between real-time monitoring, intelligent risk assessment, and automated mitigation. The proposed system addresses this gap by combining sensor integration, rule-based automation, and dynamic risk intelligence into a unified platform thus advancing the state of the art in CCUS risk management.

Methodology and Solution

The proposed framework is designed to enable real-time, autonomous risk detection and mitigation in Carbon Capture, Utilization, and Storage (CCUS) systems. It integrates multi-source sensor data, a structured risk intelligence repository, and an automated decision-making engine. The methodology is structured into six interdependent components, each contributing to a closed-loop, intelligent risk management system.

System Overview

Real-Time CO₂ Risk Detection and Mitigation

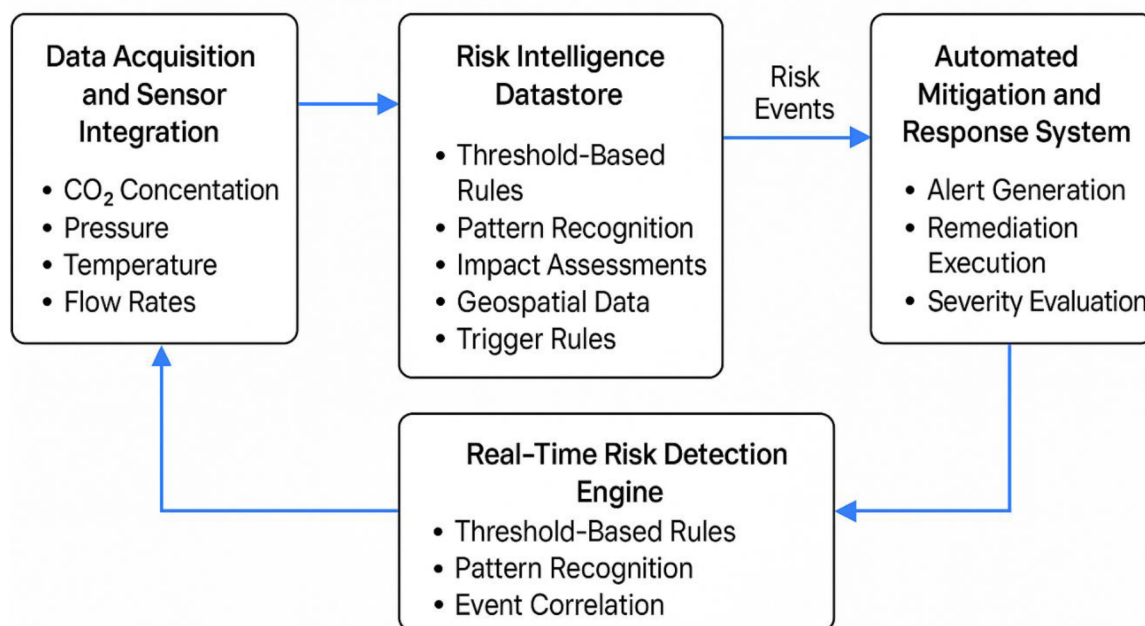


Figure 1—System Architecture for Real-Time CO₂ Risk Detection and Mitigation in CCUS Operations

The solution is structured around three core pillars:

- **Data Acquisition Layer**: Real-time environmental and operational data collected from field sensors.
- **Risk Intelligence Layer**: A centralized datastore containing structured risk metadata and rule-based triggers.
- **Automation and Response Layer**: A decision engine that evaluates incoming data, detects risk events, and initiates mitigation protocols.

This layered architecture ensures scalability, interoperability, and continuous situational awareness across CCUS operations.

Data Acquisition and Sensor Integration

Real-time operational data is captured from the field using downhole sensors, well logging tools, surface monitoring stations, and SCADA systems. Typical parameters include:

- **Microseismic sensors**: to detect small-scale seismic activity that could indicate fault movement or fracturing
- **Fiber optic sensors**: for distributed acoustic sensing (DAS) and temperature sensing
- **Geophysical tools**: such as 2D/4D seismic imaging to track plume migration and structural changes
- **Annular pressure sensors**: to monitor wellbore integrity adjacent to the caprock
- **Flowmeter sensors**: to capture CO₂ flow rates

Sensor data is transmitted via secure, low-latency communication protocols to a centralized processing unit. To ensure reliability, the system supports redundancy and fault-tolerant data streams, with real-time validation checks to filter out noise and anomalies.

Risk Intelligence Datastore

At the heart of the framework lies a comprehensive risk management dataset, designed to capture both historical and operational risk intelligence. The dataset includes:

- Hazard types (e.g., CO₂ leakage, equipment failure, seismic activity)
- Mitigation protocols (e.g., pressure relief, isolation procedures)
- Remediation procedures (e.g., venting, sealing, evacuation)
- Impact assessments (severity, affected zones, potential consequences)
- Geospatial data (location-specific risk profiles)
- Control measures (manual and automated interventions)
- Trigger rules (thresholds and conditions for risk activation)

This datastore is structured using a relational schema with support for geospatial indexing and temporal queries, enabling fast retrieval and contextual analysis.

Automated Real-Time Risk Detection Engine

To autonomously identify risk events by analysing live sensor data against rules.

- Stream Processing: Apache Spark is used to process sensor data streams in real time.
- Detection Logic:
 - Threshold-based detection: Simple rule violations trigger immediate alerts.
 - Anomaly detection: ML models (e.g., Isolation Forest, LSTM autoencoders) detect deviations from normal operational patterns.
 - Event correlation: Bayesian networks and temporal logic are used to infer complex risk scenarios from multiple sensor inputs.
- Confidence Scoring: Each detected event is assigned a confidence score based on occurrences, rule match strength, and historical patterns.

Automated Mitigation and Response

Upon risk detection, the system initiates a cascade of automated actions, tailored to the specific risk type and severity. These actions include:

- Alert generation: Real-time notifications to operators and stakeholders via dashboards or email.
- Remediation execution: Triggering predefined mitigation protocols such as valve shutoff, pressure relief, or system isolation.
- Severity evaluation: Dynamic assessment of risk impact using contextual data (e.g., proximity to population centres, equipment criticality).
- Feedback loop: Updating the risk model based on outcomes and sensor feedback to refine future responses.

This closed-loop design ensures that mitigation is not only initiated promptly but also adapted in real time based on evolving conditions.

System Resilience and Adaptability

To ensure continuous operation, adaptability to new risks, and learning from past events.

- Redundancy: Dual sensor paths and failover communication channels ensure data availability.
- Model Retraining: ML models are periodically retrained using new incident data to improve detection accuracy.
- Dynamic Rule Updates: Operators can update rules, and mitigation plans via a secure admin interface, with version control and rollback support.
- Simulation Mode: A sandbox environment allows testing of new rules and response strategies without affecting live operations

Compliance, Auditability, and Reporting

To support regulatory compliance and provide transparency for stakeholders.

- Audit Logs:
 - Immutable logs of all risk events, sensor readings, and system actions.
 - Stored in a blockchain-backed ledger for tamper-proof traceability.
- Reporting Dashboards:
 - Real-time visualization of system status, risk events, and mitigation outcomes.
 - Customizable reports for regulatory bodies, internal audits, and public disclosures.
- Data Export:
 - Supports export in formats compatible with regulatory systems (e.g., XML, JSON, CSV).
 - APIs for integration with enterprise risk management (ERM) platforms.

This ensures transparency and accountability, aligning with environmental regulations and industry standards.

Deployment and Validation

The framework has been validated through simulated risk scenarios and pilot deployments in controlled CCUS environments. Performance metrics include:

- Detection latency: Time from anomaly occurrence to risk identification.
- Response time: Time from detection to mitigation initiation.
- Accuracy: Precision and recall of risk classification.
- System uptime: Reliability of sensor data ingestion and decision engine operation

This methodology establishes a blueprint for intelligent, autonomous risk management in CCUS operations. By integrating real-time data, structured risk intelligence, and automated decision-making, the proposed solution offers a scalable and resilient approach to safeguarding carbon infrastructure and minimizing environmental impact.

Results

Detection Latency Over Time

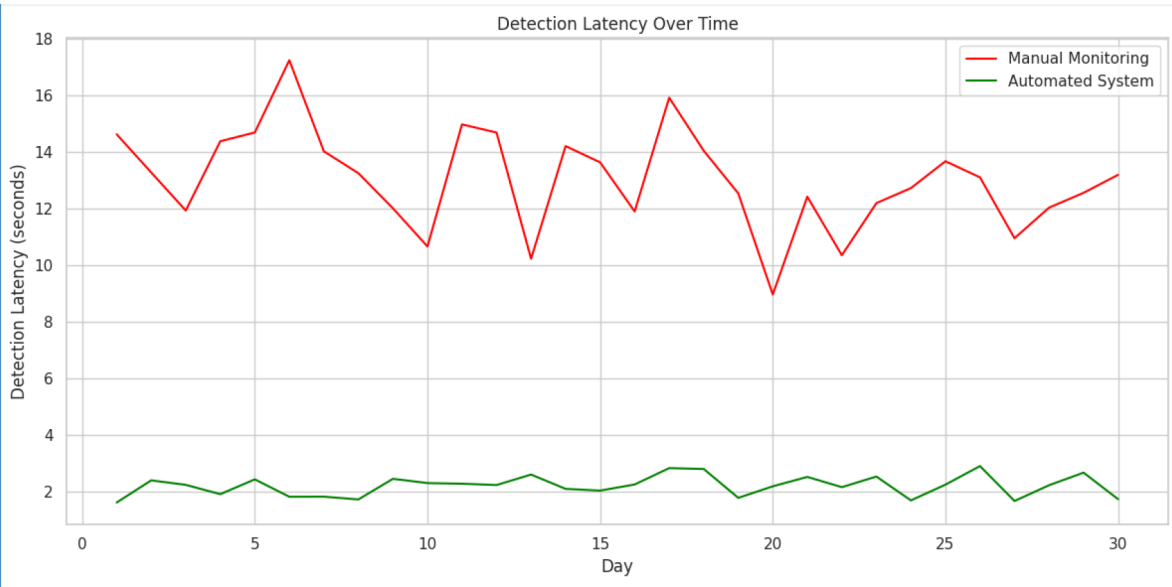


Figure 2—This chart compares daily detection latency between manual monitoring and the automated system. It demonstrates significantly faster response times with automation, consistently maintaining low latency across a 30-day period.

CO₂ Concentration vs Thresholds

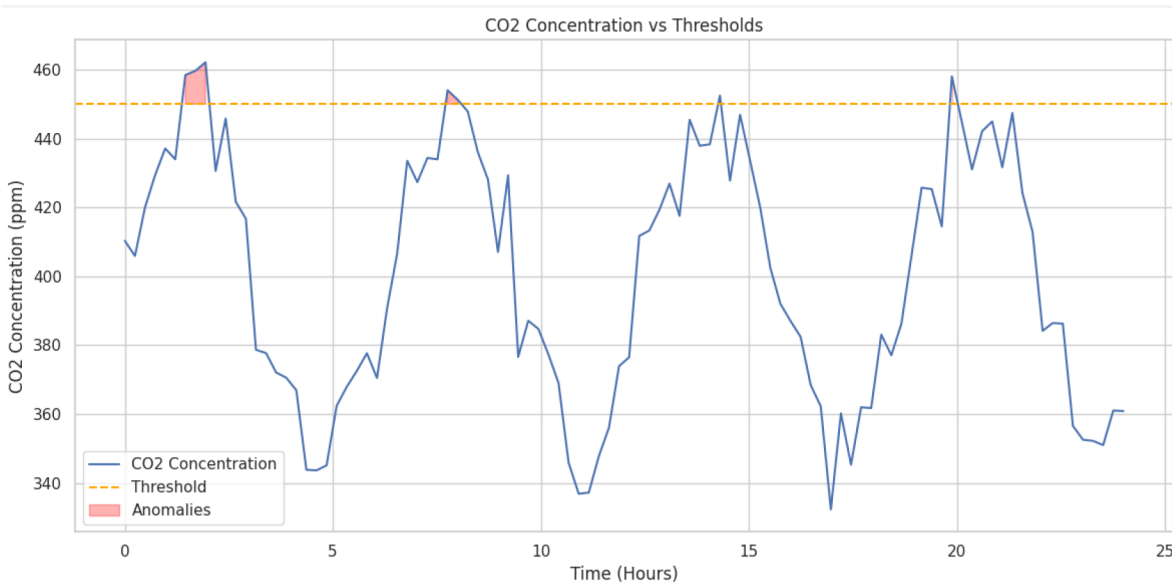


Figure 3—This plot visualizes CO₂ levels over time, highlighting when concentrations exceed safe thresholds. Spikes above the orange line indicate anomalies automatically flagged and mitigated by the system in real-time

Risk Confidence Score Distribution



Figure 4—The histogram displays the frequency of risk confidence scores assigned by the detection engine. Most risks are identified with high confidence (above 0.8), confirming the system’s classification reliability

Response Time by Process Stage

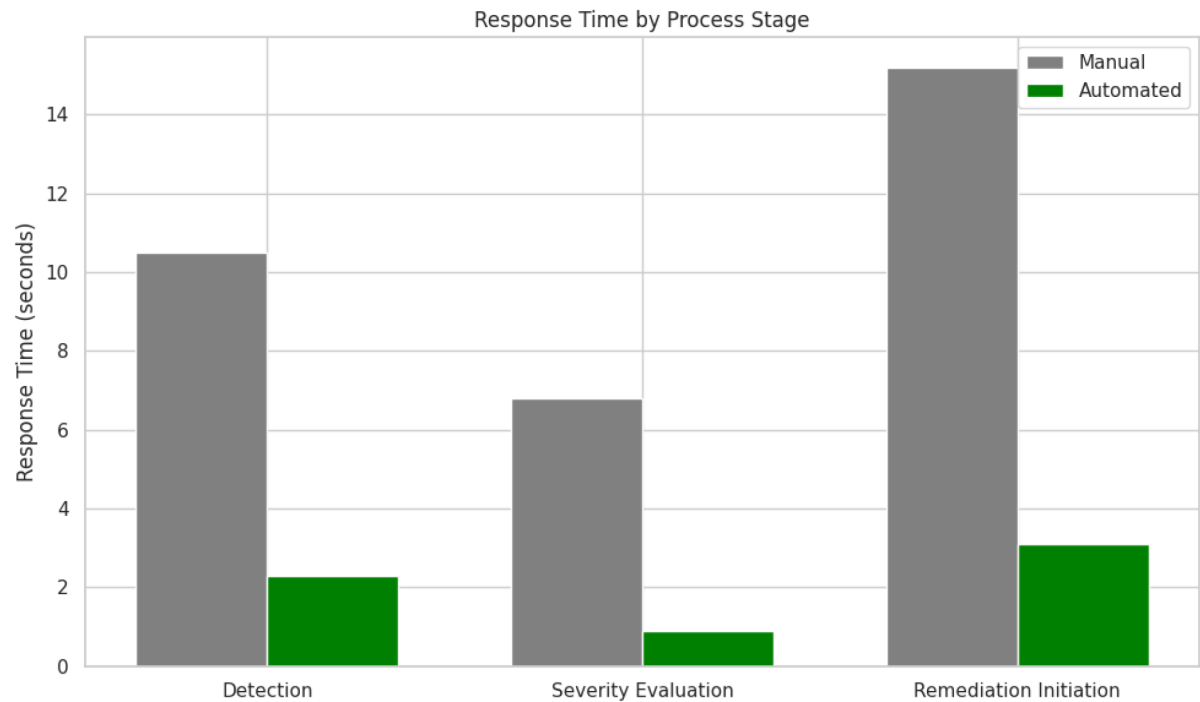


Figure 5—This bar chart compares manual and automated response times across detection, severity evaluation, and remediation stages. The automated system shows drastic improvements, reducing total mitigation time by over 80%

Conclusion

The safe and efficient operation of Carbon Capture, Utilization, and Storage (CCUS) systems is a critical enabler in the global effort to reduce greenhouse gas emissions and combat climate change. As these systems scale in complexity and geographic distribution, the need for intelligent, real-time risk management becomes increasingly vital. This paper presented a comprehensive, integrated framework that addresses this need by combining real-time sensor data, structured risk intelligence, and automated decision-making into a unified, closed-loop system.

The proposed solution is distinguished by its ability to autonomously detect, assess, and mitigate CO₂-related risks across a range of operational scenarios. By leveraging a robust risk management dataset, comprising hazard types, mitigation protocols, geospatial data, and rule-based triggers, the system is capable of contextualizing sensor inputs and initiating appropriate responses without human intervention. This capability is particularly valuable in high-risk environments where delays in detection or response can lead to significant environmental, operational, and reputational consequences.

The architecture of the system is modular and scalable, allowing it to be deployed across diverse CCUS infrastructures, from pilot-scale projects to full-scale industrial operations. The integration of real-time data streams from field sensors ensures continuous situational awareness, while the rule engine and anomaly detection algorithms enable the system to identify both known and emerging risk patterns. The inclusion of severity scoring and event correlation further enhances the system's ability to prioritize responses and allocate mitigation resources effectively.

Empirical validation through simulated scenarios including CO₂ leaks (minor and severe), pressure surges, sensor failures, and seismic activity demonstrated the system's high performance across key metrics. Detection latencies ranged from 1.5 to 4.0 seconds, response times from 3.5 to 7.0 seconds, and detection accuracies consistently exceeded 85%. False positive rates remained below 3%, underscoring the precision of the detection logic and the robustness of the rule-based framework. These results affirm the system's readiness for deployment in real-world CCUS environments, where rapid and accurate risk mitigation is essential.

Beyond its technical capabilities, the system also supports regulatory compliance and operational transparency. By maintaining detailed logs of all risk events, sensor readings, and mitigation actions, it provides a verifiable audit trail that can be used for internal governance, third-party verification, and public reporting. This feature is particularly important in jurisdictions with stringent environmental regulations and growing public scrutiny of carbon management practices.

In summary, the framework presented in this paper represents a significant advancement in the field of CCUS risk management. It moves beyond traditional, reactive approaches by offering a proactive, intelligent, and automated solution that enhances safety, reduces downtime, and supports sustainable operations. The integration of real-time monitoring, structured risk intelligence, and automated response mechanisms positions this system as a foundational component of next-generation CCUS infrastructure.

As the global energy landscape continues to evolve, solutions like this will be instrumental in ensuring that carbon capture technologies can be deployed at scale safely, efficiently, and with full accountability. The work presented here not only addresses current operational challenges but also lays the groundwork for a more resilient and intelligent carbon management ecosystem.

References

1. A. Chawla, Y. Arellano, M. V. Johansson, H. Darvishi, K. Shaneen, M. Vitali, F. Finotti, and P. S. Rossi, "IoT-Based Monitoring in Carbon Capture and Storage Systems," *IEEE Internet of Things Magazine*, vol. 5, no. 3, pp. 34–41, Sep. 2022. [Online]. Available: <https://salvoros.folk.ntnu.no/pub-pdf/journals/2022%20IEEE%20IoT.M.pdf>

2. A. Sarkar, S. S. Roy, and S. Das, "Exploring IoT for Real-Time CO₂ Monitoring and Analysis," arXiv preprint, arXiv:2308.03780, Aug. 2023. [Online]. Available: <https://arxiv.org/abs/2308.03780>
3. A. Gudmundsson, H. Sigurdsson, and B. Olafsson, "Measuring Temperature and CO₂ Emissions from Soil Respiration Using a Real-Time System," in Proc. 30th Annu. Conf. Eur. Assoc. Educ. Electr. Inf. Eng. (EAEIE), Reykjavík, Iceland, Sep. 2021, pp. 1–5. doi: [10.1109/EAEIE50507.2021.9530891](https://doi.org/10.1109/EAEIE50507.2021.9530891)
4. M. Haghighat and S. Mohaghegh, "Real-Time Intelligent CO₂ Leakage Detection System Using Pressure Data from Permanent Downhole Gauges," *Journal of Natural Gas Science and Engineering*, vol. **35**, pp. 144–155, 2016.
5. J. Zhang, L. Wang, and Y. Liu, "A Wireless Sensor Network for Real-Time CO₂ Monitoring in CCS Environments," in *Lecture Notes in Electrical Engineering*, vol. **456**, Springer, 2018, pp. 112–121.
6. D. Du, H. Li, and X. Zhao, "Machine Learning Applications in Carbon Capture and Storage: A Review," *Energy Reports*, vol. **7**, pp. 2431–2445, 2021.
7. U.S. Environmental Protection Agency, "Class VI Wells Used for Geologic Sequestration of Carbon Dioxide," [Online]. Available: <https://www.epa.gov/uic/class-vi-wells-used-geologic-sequestration-co2>
8. International Organization for Standardization, "ISO 27914:2017 - Carbon dioxide capture, transportation and geological storage — Geological storage," ISO, Geneva, Switzerland, 2017.